

JiffyDOS

Manual

Using JiffyDOS 6.01 (C64 kernal) and JiffyDOS 5.0 (1541)

with notes on use with the 1541 Multi-Speeder V3 and DolphinDos26

new edition based on the JiffyDOS V6.01 User's Manual

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1 Introduction

1.1 What is JiffyDOS?

JiffyDOS is an enhanced operating system for the Commodore C64, SX-64 and C128 and their disk drives. It was developed by **Mark Fellows** and sold as version 6 by **Creative Micro Designs (CMD)** from 1989; today JiffyDOS is licensed and sold by **Jim Brain, RETRO Innovations**.

JiffyDOS consists of two parts: a kernal ROM for the computer and a DOS ROM for each drive. Both replace the original ROMs and can be switched back to the Commodore ROMs. Because JiffyDOS lives in ROM, it needs no port, no extra cable and no memory in the computer – so it speeds up almost any software that uses the kernal routines to load, save and read files.

- Loading about **ten times faster** than with the original DOS; saving and reading SEQ, REL and USR files much faster.
- The complete **DOS Wedge 5.1 command set** (@, /, %, ← …) is built in, plus many commands of its own.
- Preset **function keys** that work together with the directory: pick a file, press the key, press RETURN.
- Show the directory without destroying the BASIC program in memory; pause listings and step through them line by line.
- Built-in **file copier** between two drives, adjustable **sector interleave**, screen dump, un-NEW.
- DOS error checking is kept; disks and files stay in standard format.



Figure 1: Start-up screen of the JiffyDOS 6.01 kernal

1.2 About this manual

This manual is a **new edition** of the “JiffyDOS V6.01 User's Manual” by Creative Micro Designs, written in my own words and extended with notes on use in the DolphinDos26 project. It is **not an official manual** from CMD or RETRO Innovations. Installation instructions for the individual computers and drives are not included; they come with the JiffyDOS ROMs.

It describes JiffyDOS on the **C64** (or C128 in 64 mode) with a **1541**. Differences for C128 mode and the 1571 and 1581 drives are mentioned where they apply. Separate chapters cover use with SD2IEC and with Pi1541.

Notation: input is shown in typewriter font, e.g. `LOAD`. `name` stands for a file name, `g` for a device number (4 or 5 for printers, 8 to 11 for drives). Anything in square brackets `[ ]` is optional; the brackets themselves are not typed. `…` means the last item may be repeated. A command may be at most **41 characters** long. Keys are shown in angle brackets, e.g. `<CTRL>+<D>` (hold CTRL and press D).

Some commands use special characters of the C64 keyboard: `↑` (up arrow, right of the asterisk), `←` (left arrow, top left) and `£` (pound, right of the minus).

1.3 JiffyDOS in the DolphinDos26 project

On the **1541 Switchless Multi-Speeder V3**, JiffyDOS is a drive ROM in its own bank of the EPROM. You switch with a DOS command, and the drive restarts afterwards: `@I:4@RNROM` switches to JiffyDOS 5.0 (with any kernal: `OPEN 1,8,15,"I:4@RNROM":CLOSE 1`), `@I:8@RNROM` switches back to DolphinDos26.

Bank	Content	Meaning for JiffyDOS
4	JiffyDOS 5.0 (1541)	the JiffyDOS drive ROM that belongs to JiffyDOS 6.01 (see “ROM switching”)
5	S-JiffyDOS	JiffyDOS variant with track cache; not covered by this manual
8	DolphinDos26	does not speak the JiffyDOS protocol; with a JiffyDOS kernal it runs serially at normal speed

There are two choices for the computer:

C64 kernal	Drive	Result
Original JiffyDOS kernal 6.01	JiffyDOS (bank 4)	full JiffyDOS speed, all commands and features of this manual
Original JiffyDOS kernal 6.01	DolphinDos26 (bank 8) or CBM DOS	all commands of this manual, but only normal serial speed; no parallel cable
`C64DolphinDos26.bin`	JiffyDOS (bank 4), SD2IEC	fast transfer using the JiffyDOS protocol (shown as `LOADING JF:`); the commands are those of DolphinDOS, not those of this manual
`C64DolphinDos26.bin`	DolphinDos26 (bank 8)	parallel transfer over the DolphinDOS cable (see the DolphinDos26 manual)

The C64 kernal `C64DolphinDos26.bin` contains only the JiffyDOS **transfer routines**, not its commands, function keys and copier. To use the commands in this manual you need the original JiffyDOS kernal, e.g. in a kernal switch next to `C64DolphinDos26.bin`. More on this in the chapter “Using the DolphinDos26 kernal”.

C128 and 1571: The C128 kernals `C128DolphinDos26-US.BIN` and `C128DolphinDos26-DE.BIN` also contain the JiffyDOS transfer routines, in C128 mode with block-wise fast loading into any memory bank (display `LOADING JF:`). On the **1571 Multi-Speeder V3** in the C128DCR, JiffyDOS 6.0 is in banks 5 to 8 (`@I:5@RNROM`), DolphinDos26 for the 1571 in bank 4 (`@I:4@RNROM`); like DolphinDos26 for the 1541, this firmware does not speak the JiffyDOS protocol. With the JiffyDOS

1571 ROM, the C64 kernal `C64DolphinDos26.bin` also transfers using the JiffyDOS protocol in the C64 mode of the C128. Both work on PAL and NTSC computers.

> **Tested:** Switching to JiffyDOS 5.0 in bank 4, and LOAD and SAVE using the JiffyDOS protocol with the C64 kernal DolphinDos26, have been checked on a real Multi-Speeder V3. The figures in this manual show the original JiffyDOS kernal 6.01 with the JiffyDOS 1541 ROM.

1.4 Where to get it, and licensing

JiffyDOS is **commercial software** and is still available. ROMs and licences are sold by RETRO Innovations: www.jiffydos.com. This manual and the DolphinDos26 project contain no JiffyDOS ROMs for download. Please check for yourself whether you need a JiffyDOS licence for your use (see the appendix “Copyright and terms of use”).

2 Features

2.1 Characteristics

Feature	Description
No port, no cable	JiffyDOS uses only the serial bus. Expansion, user and cassette ports stay free – modems, memory expansions and other hardware are not affected. (Systems such as DolphinDOS, SpeedDOS or RapiDOS need a parallel cable instead.)
DOS Wedge built in	The complete command set of the Commodore DOS Wedge 5.1, extended with command chaining, string variables and a device number at the end of a command.
Additional commands	Turn off head knocking, lock files, list BASIC and text files directly from disk, un-NEW, screen dump, file copier, interleave.
Convenience	Directory without losing memory, load and run the first program with one key, pause listings, function keys with commands.
Full error checking	All read and write operations keep the DOS checks; JiffyDOS does not trade away safety (unlike fast formatters with 10-second formats).
Current ROM versions	The JiffyDOS ROMs are based on the latest Commodore versions; the 1571 ROM has bugs fixed (REL files, initialisation).

2.2 Speed

JiffyDOS speeds up every drive operation: loading (including copy-protected software), saving, reading and writing files from within programs, scratching, validating and, on the 1541, formatting. Because JiffyDOS changes no vectors and uses no memory, it works equally in word processors, databases, programming languages, MIDI software and games – as long as they use the kernal routines. This applies to all file types (PRG, SEQ, REL, USR).

The tables give the times in seconds from the original manual, rounded, without motor spin-up (about half a second) and without the directory search. “Std.” is the original Commodore system, “JD” is JiffyDOS.

2.2.1 C64, SX-64 and C128 in 64 mode

Operation	1541 Std.	1541 JD	1571 Std.	1571 JD	1581 Std.	1581 JD
Load a 202-block program	124	12	124	9	102	8
Save a 100-block program	75	24	75	20	40	15
Read a 125-block SEQ/USR file	84	15	84	13	63	9
Write a 100-block SEQ/USR file	81	27	81	24	44	17
Read 64 REL records of 154 bytes	40	14	40	14	37	10
Write one REL record of 154 bytes	0.350	0.125	0.350	0.120	0.325	0.110
16 KB over the command channel	47	9	47	9	47	9

2.2.2 C128 in 128 mode

Operation	1541 Std.	1541 JD	1571 Std.	1571 JD	1581 Std.	1581 JD
Load a 202-block program	124	12	14	9	12	8
Save a 100-block program	75	24	48	25	26	14
Read a 125-block SEQ/USR file	84	15	31	12	20	10
Write a 100-block SEQ/USR file	81	27	48	33	20	11
Read 64 REL records of 154 bytes	40	14	21	14	17	10
Autoboot a 202-block program	125	13	54	10	13	9
16 KB over the command channel	47	10	10	6	10	6

How much of this you see in practice depends on the software and on the interleave of the files – see “Getting maximum speed” and “If it does not get faster”.

2.3 Compatibility

- **Software:** JiffyDOS loads practically all programs, including many copy-protected ones that do not run with cartridges or turbo loaders. Programs with their own fast loader (e.g. GEOS) keep working – but they are then not sped up by JiffyDOS.
- **Hardware:** Because no port is used, JiffyDOS works with modems, RAM expansions, MIDI interfaces and hard disks. Drives and printers without JiffyDOS also work on the serial bus.
- **Mixed systems:** A JiffyDOS drive works on an original computer and vice versa – each at normal speed. Drives of different types (1541, 1571, 1581), with or without JiffyDOS, can share the bus; the device number can be set freely.
- **Standard format:** Disks and files written with JiffyDOS are Commodore standard and readable on any other system.
- **Can be switched off:** If a program does not run, a switch (on the Multi-Speeder a command) returns to the original ROMs – even with the power on.
- **RAMDOS:** The JiffyDOS commands work with Commodore's RAMDOS for the 1700, 1750 and 1764 REUs and with the RAM DOS of the CMD RAMLink.

2.4 Supported devices

Group	Devices
Computers	C64, C64 II (“64C”), SX-64, C128, C128D
Commodore drives	1541, 1541C, 1541-II, 1571, 1581
Compatible drives	FSD-1, FSD-2, Excelerator+, Excel 2001, OC-118, MSD SD-1 and SD-2, Enhancer 2000, Blue Chip BCD/5.25 and BCD/128, RF501C, RF512C, FD-168, SW71, Indus GT, CSD-1

Each drive that is to be sped up needs its own JiffyDOS ROM.

3 Operation

3.1 ROM switching

Usually JiffyDOS can stay switched on all the time. Only in rare cases do you need to switch back to the original ROMs for a program.

3.1.1 C64, SX-64 and C128

Switch the computer off, flip the kernal switch, switch it on again. The start-up screen shows which kernal is active; with JiffyDOS the first line reads `JIFFYDOS V6.01 (C)1989 CMD`, with the original kernal the usual BASIC message appears.

3.1.2 1541 and compatible drives

Switch the drive off, flip the switch, switch it on again (the SX-64 has no separate drive switch). The error channel right after power-up shows which ROM is active – with JiffyDOS, simply type `@`:

```
73,JIFFYDOS 5.0 1541,00,00
73,CBM DOS V2.6 1541,00,00
```

The first message means JiffyDOS, the second the original DOS. The drive ROM reports version 5.0 because it has not changed since version 5; it belongs to JiffyDOS 6.01.

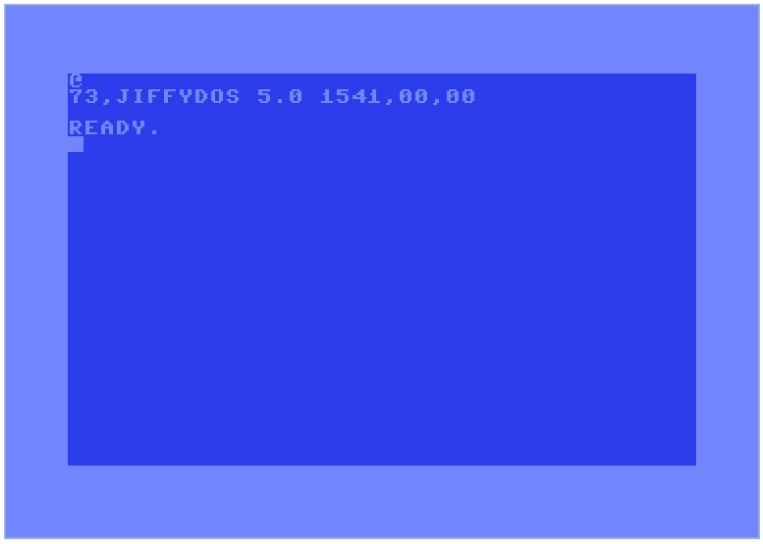


Figure 2: Error channel after power-up: JiffyDOS 1541 ROM active

3.1.3 1541 with Multi-Speeder V3

Instead of a switch there are the DOS commands from the chapter “JiffyDOS in the DolphinDos26 project”: `@I:4@RNROM` for JiffyDOS, `@I:1@RNROM` for CBM DOS, `@I:8@RNROM` for DolphinDos26. The LED blinks n times, then the drive restarts; after a few seconds it is ready again.

3.1.4 1571 and 1581

These drives switch in software: they detect whether the computer runs the JiffyDOS or the original kernal and choose the matching routines. No switch is needed. With the 1571 Multi-Speeder V3 in the C128DCR, the drive ROM is selected with `@I:n@RNROM` as on the 1541.

3.1.5 Switching with the power on

The ROMs may be switched during operation – useful when a heavily copy-protected program only loads without JiffyDOS but then runs faster with JiffyDOS. The only condition: **no drive access** may be in progress, or the system hangs. Switching the kernal in the middle of a running program is not always reliable; some programs crash.

- **From JiffyDOS to the original kernal (in BASIC direct mode):** first type `@Q`, then flip the switch.
- **From the original kernal to JiffyDOS (in BASIC direct mode):** flip the switch, then type `SYS 58551` (64 mode) or `SYS 65137` (128 mode).

3.2 Datasette

No datasette can be used with the JiffyDOS kernal; accessing device 1 ends with `?ILLEGAL DEVICE NUMBER ERROR`. For tapes, switch to the original kernal.

3.3 Function keys

In BASIC direct mode the function keys carry JiffyDOS commands. As soon as a program runs, JiffyDOS turns this off so that it does not interfere with the program's own keys.

Key	Assignment	Effect
F1	`@$:*` and RETURN	Show the directory
F3	`/`	Load a BASIC program
F5	`↑`	Load and run a BASIC program
F7	`%`	Load a machine-language program
F2	`@D`	List a BASIC program from disk
F4	`@T`	List a text (ASCII) file from disk
F6	`←`	Save a BASIC program
F8	`@ "S:`	Scratch a file

3.3.1 Function keys with the directory

The keys are designed for the lines of the directory: show the directory with F1, move the cursor to the line of the file, press the function key and confirm with RETURN. File type and block count do not get in the way, no colon is needed, and typing errors in the file name are impossible. The name can still be changed before RETURN.



Figure 3: Directory with @$, then loading a program with /

3.3.2 Turning them off and on

`@F` turns off the JiffyDOS assignment (in 128 mode the normal C128 assignment then applies). `SYS 58551` (64 mode) or `SYS 65137` (128 mode) restores it.

3.3.3 Your own assignment (64 mode)

The keys can be reassigned with a small BASIC program. The texts are in F1$ to F8$; they must be joined in the order F1, F3, F5, F7, F2, F4, F6, F8, and each ends with CHR$(0). The program stores them in the cassette buffer (address 820) and sets the pointer in addresses 176/177; POKE 155,0 turns the assignment on.

```
10 F1$="@$:*"+CHR$(13)+CHR$(0)
20 F2$="@D"+CHR$(0)
30 F3$="/"+CHR$(0)
40 F4$="@T"+CHR$(0)
50 F5$="↑"+CHR$(0)
60 F6$="←"+CHR$(0)
70 F7$="%"+CHR$(0)
80 F8$="@  "+CHR$(34)+"S:"+CHR$(0)
90 F$=F1$+F3$+F5$+F7$+F2$+F4$+F6$+F8$
100 BUF=820
110 FOR I=1 TO LEN(F$)
120 POKE BUF+I-1,ASC(MID$(F$,I,1)):NEXT I
130 POKE 176,BUF-256*INT(BUF/256)
140 POKE 177,INT(BUF/256):POKE 155,0
```

BUF may point to any free RAM area, just not under the BASIC or kernal ROM.

3.4 Pausing listings

Long listings – BASIC LIST, directory with `@$`, `@D` and `@T` – can be paused and stepped through line by line. This also works in many commercial programs.

Key	Effect
Hold `<CTRL>`	The listing stops while the key is held

Key	Effect
<CTRL>+<S>	Pause the listing; you can take your hands off the keyboard. Continue with any key except CTRL, RUN/STOP, SHIFT and C=.
Hold <CTRL>, alternately S and W	Step line by line: each W shows a new line; releasing CTRL continues normally
Hold <RUN/STOP>	Abort the listing (also while paused; not always in other programs)

3.5 Getting maximum speed

With the 1541 and 1571, files saved under JiffyDOS load faster than files from the original DOS – sometimes up to three times as fast. JiffyDOS lays out the blocks on the disk so that it finds the next block sooner (see “Sector interleave”). The content and format of the file do not change; it stays readable on original systems.

- **Programs:** load once with JiffyDOS and save again.
- **Files of other programs** (texts, MIDI songs …): load in the program concerned and save again; often twice as fast the next time.
- **Whole disks:** copy the files with the JiffyDOS copier to an empty disk – this also removes fragmentation.
- **Special cases:** if that does not help, adjust the interleave (see below).

3.6 The file copier

JiffyDOS has a built-in copier for two drives. It is not the fastest, but it has many advantages:

- built in, no program to load; the BASIC program in memory is kept when copying single files
- copies PRG, SEQ, REL and most USR files
- replaces files of the same name on the destination automatically (scratch first, then write – no “save with replace”)
- writes with the JiffyDOS interleave or the value chosen with @G
- works between any Commodore-compatible drives, even without JiffyDOS, and with RAMDOS REUs
- usable in your own BASIC programs; can change the file type of PRG, SEQ and USR

Command	Effect
@Xg	Set the destination drive for copying
*"name"typ	Copy a file to the destination drive (typ = PRG, SEQ, REL or USR)
<CTRL>+<A>	Toggle the copy mark of all files in the loaded directory
<CTRL>+<W>	Toggle the copy mark of the file in the cursor line

3.6.1 Copying single files

1. Put the source disk in the source drive and the destination disk in the destination drive (not needed for an REU).
2. Set the destination, e.g. @X9 for device 9.
3. Choose the source as default device: @#8 or <CTRL>+<D>.
4. Show the directory with @$ or F1; stop it with RUN/STOP when the file is visible.

5. Move the cursor to the line of the file, stay at the far left and press `*` – the asterisk appears in the first column.
6. Press RETURN: the file is copied. For more files repeat from step 4.

3.6.2 Copying several files

1. Steps 1 to 3 as above.
2. Load the directory as a program with `/$` (this overwrites the BASIC program in memory) and show it with `LIST`.
3. Mark files: `<CTRL>+<A>` toggles the mark of all files and lists again; `<CTRL>+<W>` toggles the file in the cursor line (cursor at the far left). Marked files carry an asterisk.
4. Type `RUN`: the copier works through the marked files; then the cursor returns.

> **Important:** To copy several files, always load the directory with `/$`, do not show it with `@$` – otherwise `<CTRL>+<A>` and `<CTRL>+<W>` do not work. And never press RETURN on a line with a file name: this destroys the file list, and the directory must be loaded again.

3.6.3 Copying in BASIC programs

The first example copies the PRG file JIFFYMON and the SEQ file TAX DATA from device 8 to device 10. The second asks for source, destination, file name and type and copies the file accordingly.

```
100 @#8
110 @X10
120 *"JIFFYMON"PRG
130 *"TAX DATA"SEQ
140 END
```

```
100 INPUT"SOURCE DRIVE";S
110 INPUT"DESTINATION";D
120 @#S:@XD
130 INPUT"FILENAME";F$
140 INPUT"TYPE (P/S/R/U)";T$
150 IF T$="P" THEN *F$ PRG
160 IF T$="S" THEN *F$ SEQ
170 IF T$="R" THEN *F$ REL
180 IF T$="U" THEN *F$ USR
```

3.6.4 Changing the file type

Show the directory as for copying a single file and press `*` on the line of the file, but do not press RETURN yet. On the same line overwrite the type (PRG, SEQ, USR) with the new type, then press RETURN: the copy gets the new type. In programs this works the same way with `*F$ SEQ` etc. REL files can neither be turned into another type nor created from another type.

3.6.5 Notes on the copier

- Special files such as 1581 partitions (“CBM”) and the USR files of GEOS cannot be copied.
- When changing between PRG and SEQ/USR, the first two bytes (load address) stay as they are: they are neither removed nor added.
- Before copying a REL file, JiffyDOS determines the record length; this can take up to 40 seconds while the destination drive is idle – the system has not hung.

- After copying a REL file the error LED of the source drive blinks; this is not an error.
- For files in 1581 partitions, first change into the partition with `@/0:name` (see appendix).

3.7 Sector interleave

The interleave is the distance between two consecutive blocks of a file on a track. If it is too small, the next block has already passed and the drive waits a whole revolution (0.2 seconds). If it is too large, it also waits unnecessarily. JiffyDOS transfers faster than the original DOS and therefore uses its own values: for PRG files the fastest value for the kernal LOAD, for SEQ, REL and USR a compromise.

Drive	JiffyDOS: PRG	JiffyDOS: SEQ, REL, USR	Original DOS: all types
1541	6	10–12 (depending on track zone)	10
1571	4	8–10 (depending on track zone)	6

The standard values fit almost always. A different interleave only helps with files that are not read via the kernal LOAD:

- programs with their own loader that reads the file byte by byte,
- PRG data files that a program reads without the kernal LOAD,
- SEQ, REL and USR files that a program reads more slowly than usual (e.g. compiled BASIC).

If the times are already close to the values in the tables in chapter 2, a different interleave does not help – it is more likely to make things worse. If a program loads two to three times slower than possible, it is worth a try.

3.7.1 Finding the best value

1. Provide two drives: source (original) and destination. Choose a starting value: 6 for the 1541, 4 for the 1571.
2. Make a copy of the original disk – for program disks an exact copy (nibble, fast or parameter copy), for data disks only with the system files, drivers etc., without the actual data.
3. Put the copy in the destination drive and the original in the source drive.
4. Choose the destination as default device (`@#9` or `<CTRL>+<D>`) and set the interleave, e.g. `@G5`.
5. Choose the source as default device (`@#8`), load the directory with `/$`, `LIST`, mark the files concerned with `<CTRL>+<A>` / `<CTRL>+<W>`.
6. Set the destination with `@X9` and copy with `RUN`.
7. Test the destination disk: load the program or the data and time it; note the result.
8. Increase the interleave by one and repeat from step 3, up to and including 16.
9. Choose the best value and copy the files with it for good.
10. Afterwards restore the standard interleave on all drives – most safely by switching off and on (C128: reset button) or with `@G0`.

3.7.2 Working with data files that need a different interleave

1. Before starting the program, choose the data drive as default device, e.g. `@#9`.
2. Set the interleave found, e.g. `@G9` (1541 and 1571 only; no effect on the 1581). With several data drives repeat for each.
3. Choose the drive the program loads from as default device again, e.g. `@#8`.

4. Load and run the program, read and write data as usual.
5. Afterwards restore the standard interleave.

3.7.3 Important points about the interleave

- Effective only on the 1541, 1571 and closely compatible drives (FSD, Blue Chip, Excel, Enhancer and others). On MSD and Indus drives a changed interleave can disturb the DOS and damage data – do not use it there.
- The right value depends on many factors; usually only trial and error helps.
- The interleave does not make a slow program fast. BASIC with GET# in particular stays slow.
- Programs with their own access routines that bypass the kernal (e.g. GEOS) benefit neither from JiffyDOS nor from the interleave.
- Files saved with the JiffyDOS interleave usually load more slowly on original systems.
- Write disks used on both 1541 and 1571 with the best value for the slower 1541.

3.8 If it does not get faster

Cause	Explanation and remedy
Own fast loader	Programs such as GEOS, some MIDI sequencers, copy programs or disks created with freezer cartridges bring their own DOS and bypass the kernal and drive ROM – JiffyDOS has no effect. The only remedy is a program version without a fast loader, or a different interleave.
BASIC programs	The interpreter needs a lot of time for each statement. `LOAD"TEST.ML",8,1` from BASIC runs at full speed; reading byte by byte with GET# gives at most about a factor of 2. Use INPUT# with strings as long as possible instead. Compiled BASIC is faster but likewise depends on the programming style.
Machine-language programs	Only well-written routines reach full speed; performance varies from program to program.
Fragmented files	After many scratched files, new files are scattered over the disk. Remedy: copy the files to an empty disk.
Sprites	The VIC-II halts the CPU for sprites; this disturbs the time-critical transfer. JiffyDOS therefore turns the sprites off for each byte and becomes as slow as the original system. The only remedy is in the program itself (turn sprites off during drive access, as GEOS does).
1571 on the C64	Full speed in 64 mode only in the 1541 mode of the 1571. Do not switch to 1571 mode with `U0>M1`; on the C128 do not use `GO64` but reset with the C= key held to enter 64 mode. After `GO64`, `U0>M0` helps. With JiffyDOS the 1571 reads and formats both sides even in 1541 mode.

3.9 If a program does not load or run

Loading problems almost always come from elaborate copy protection that relies on the head step rate, serial bus timing or particular ROM versions (the same schemes also cause trouble on the 1571 and 1541C). Rarely, a program loads but does not run because it calls ROM routines at unofficial entry points. In both cases switch to the original ROMs – on the Multi-Speeder with `@I:1@RNROM` to CBM DOS.

3.10 RAM expansions (RAMDOS)

The JiffyDOS commands support Commodore's RAMDOS for the 1700, 1750 and 1764 REUs in 64 and 128 mode. RAMDOS **4.2** (64 mode) or **4.3** (128 mode) or newer is required; older versions contain bugs. RAMDOS is public domain.

1. Switch on or reset the computer with JiffyDOS active.
2. Start RAMDOS as usual – but do **not** load the DOS Wedge from the RAMDOS disk; it turns off the JiffyDOS commands.
3. Access the REU with the JiffyDOS commands.

Do not use with RAMDOS: `@L` (lock file) and `£` (load and run a machine-language program). RAMDOS depends on software vectors that many programs reset; JiffyDOS cannot change that either.

3.11 Memory use in BASIC

For each command JiffyDOS sets up a 255-byte buffer at the top of BASIC memory. If a BASIC program fills almost all of memory, `?OUT OF MEMORY ERROR` can occur. Remedy: type `CLR` in direct mode, or make the program or its variables and arrays smaller.

4 The JiffyDOS commands

JiffyDOS contains the complete command set of the Commodore DOS Wedge 5.1 and additional commands of its own. All commands work in direct mode and in BASIC programs. For each command the form in **direct mode** and in **program mode** is given. The program form usually also works directly and is needed when a device other than the default device is to be addressed (`,g` at the end).

With dual drives (e.g. MSD SD-2, CBM 4040) the drive number 0 or 1 goes directly before the colon, e.g. `@$1:*` or `@S0:name`. On the 1541 it can be left out.

4.1 Setting the default device

Direct mode	Program mode
`@#g`	`@#g`

Sets the drive that all JiffyDOS commands go to. JiffyDOS checks whether the device answers; if not, `?DEVICE NOT PRESENT ERROR` appears and the previous default device remains.

Example	Effect
`@#9`	Device 9 becomes the default device (if present)

`<CTRL>+<D>` conveniently switches to the next drive present in direct mode and shows the new number – e.g. from 8 to 9 and back to 8.

4.2 Showing the directory

Direct mode	Program mode
`@$[:muster]`	`@"$[:muster]"[,g]`

Shows the directory without overwriting the program in memory. With a pattern (muster) only matching entries are shown; `*` and `?` are allowed as wildcards. The listing can be paused (see “Pausing listings”) and redirected to the printer with `@P`.

Example	Effect
`@$`	whole directory
`@$:TEST`	only the file TEST, if present
`@$:T*`	all files beginning with T
`@P`, `@$`, `@P`	print the directory, then output goes back to the screen

4.3 Reading the error channel

Direct mode	Program mode
`@`	`@""[,g]`

Reads the drive's error channel and shows the message – this replaces the little program otherwise needed and also stops the error LED from blinking. Without an error `00, OK,00,00` appears; right after power-up, the drive's DOS version.

4.4 Loading BASIC programs

Direct mode	Program mode
`/name`	`/"name"[,g]`

Loads a BASIC program like `LOAD"name",g` from the default device.

Direct mode	Program mode
`↑name`	`↑"name"[,g]`

Loads a BASIC program and runs it (like `LOAD` and `RUN`).

4.5 Saving BASIC programs

Direct mode	Program mode
`←name`	`←"name"[,g]`

Saves the BASIC program like `SAVE"name",g` to the default device.

4.6 Loading machine-language programs

Direct mode	Program mode
`%name`	`%"name"[,g]`

Loads a file to its load address like `LOAD"name",g,1`, but does **not** reset the BASIC pointers. So no `?OUT OF MEMORY` errors occur, and a BASIC program continues with the next statement after loading. This lets you load several machine-language parts from a program:

```
10 %"ML1":%"ML2":%"ML3"
```

Direct mode	Program mode
`£name`	`£"name"[,g]`

Loads a machine-language program and starts it at its load address – no `SYS` needed. Does not work with programs whose start address differs from the load address.

4.7 Loading and running the first program

`<SHIFT>+<RUN/STOP>` loads the first program on the disk with `LOAD":*",8,1` and starts it with `RUN`. In 64 mode the C= key works instead of SHIFT. Ideal for starting most commercial programs.

4.8 Verifying programs

Direct mode	Program mode
`'name`	`'"name"[,g]`

Compares memory with the file on disk like `VERIFY"name",g,1`. Works with all machine-language programs and with BASIC programs saved in the same BASIC (a BASIC program saved on the C64 cannot be verified in 128 mode).

4.9 Listing a BASIC program from disk

Direct mode	Program mode
`@D:name`	`@D"name"[,g]`

Lists a BASIC program directly from disk without loading it – to the screen or, with `@P`, to the printer. Use only for BASIC programs. The listing can be paused.

Example	Effect
`@D:GAME`	list the program GAME from disk to the screen
`@P`, `@D:LOTTERY`, `@P`	list the program LOTTERY to the printer (device 4), then back to the screen

4.10 Listing a text file from disk

Direct mode	Program mode
`@T:name`	`@T"name"[,g]`

Outputs an ASCII text file (e.g. from a word processor) to the screen or, with `@P`, to the printer. Other files produce unreadable characters. The listing can be paused.

Example	Effect
`@T:LETTER`	file LETTER to the screen
`@P`, `@T:APPOINTMENTS`, `@P`	print the file APPOINTMENTS, then back to the screen

4.11 Recovering a deleted BASIC program

Direct mode	Program mode
`@O`	–

“Un-NEW”: recovers a BASIC program lost by `NEW` or a reset – provided no new line has been entered since.

4.12 Initialising the drive

Direct mode	Program mode
`@I`	`@"I"[,g]`

Reads the BAM and ID of the inserted disk into the drive, clears the error channel and turns off the blinking error LED. Needed after a `DRIVE NOT READY`, recommended after every disk change. Read the error channel with `@` first if you still need the message.

4.13 Resetting the drive

Direct mode	Program mode
`@UJ`	`@"UJ"[,g]`

Resets the drive. Older 1541 drives may need `@U:` instead.

4.14 Validating a disk

Direct mode	Program mode
`@V`	`@"V"[,g]`

Rebuilds the BAM and frees blocks taken by files that were not closed and then scratched. Takes a while depending on the number and size of the files.

4.15 Formatting disks

JiffyDOS checks more strictly when formatting than the original DOS. If it reports an error, the disk is probably not reliable.

Direct mode	Program mode
`@N:diskname,id`	`@"N:diskname,id"[,g]`

Full format: needed for new disks. The name has up to 16 characters, the ID two characters; it is stored in the directory and in every block header.

Direct mode	Program mode
`@N:diskname`	`@"N:diskname"[,g]`

Short format: only clears the directory and BAM of an already formatted disk and is much quicker.

Direct mode	Program mode
`@N2:diskname,id`	`@"N2:diskname,id"[,g]`

Both sides on the 1571 in 1541 mode (only 1571 with JiffyDOS): formats both sides although the 1571 runs in 1541 mode in 64 mode. On other drives the command gives a syntax error and nothing is formatted.

Example	Effect
`@N:NEWDISK,00`	full format, name NEWDISK, ID 00
`@N:BACKUP`	short format, new name BACKUP
`@N2:TWOSIDE,01`	format both sides of a 1571 disk in 1541 mode
`@"N:NEWDISK,00",9`	full format of the disk in device 9

4.16 Turning off 1541 head knocking

Direct mode	Program mode
`@B`	`@B`

JiffyDOS/64 only: turns off the head rattling on read errors. `@UJ` turns it back on. At the start of a format the knock is kept so that formatting is accurate. A device number at the end is not possible.

4.17 Copying files

Direct mode	Program mode
`@Xg`	`@Xg`

Sets the destination drive: any drive on the serial bus, or the device number of an REU.

Direct mode	Program mode
`*"name"typ`	`*"name"typ`

Copies the file to the destination chosen with `@X`; typ is PRG, SEQ, REL or USR.

Example	Effect
`@X10`	destination is device 10
`*"TEST"PRG`	copy the PRG file TEST to the destination

<CTRL>+<A> toggles the copy mark of all files of a directory loaded with `/$` and lists it again. **<CTRL>+<W>** toggles the mark of the file in the cursor line (cursor at the far left). Marked files carry an asterisk; `RUN` starts copying. For the procedure see “The file copier”.

4.18 Changing the sector interleave

Direct mode	Program mode
`@Gwert`	`@Gwert`

Sets the interleave (wert = value) of the default drive for all following writes. 0 to 16 are allowed; `@G0` restores the JiffyDOS standard. The value is **not checked** – larger values cause unpredictable behaviour when writing; then simply repeat the command with a valid value. Only for the 1541, 1571 and compatible drives, not for the 1581 or MSD. A device number at the end is not possible.

Example	Effect
`@G8`	interleave of the default drive set to 8
`@G0`	restore the JiffyDOS standard

4.19 Duplicating and combining files

Direct mode	Program mode
`@C:neu=alt[,alt2]…`	`@"C:neu=alt[,alt2]…"[,g]`

Creates a copy of a file (alt = old) under a new name (neu = new) on the **same** disk, or combines several text or data files into one (programs cannot be joined this way). To copy to another disk, use the file copier.

Example	Effect
`@C:BACKUP=ORIGINAL`	create BACKUP as an exact copy of ORIGINAL
`@C:DATA=BITS,BYTES`	build DATA from BITS and BYTES

4.20 Renaming files

Direct mode	Program mode
`@R:neu=alt`	`@"R:neu=alt"[,g]`

Renames a file. The new name is on the left, the old one on the right of the equals sign.

Example	Effect
`@R:REPORT=NOTES`	NOTES is called REPORT afterwards

4.21 Scratching files

Direct mode	Program mode
`@S:name[,name2]…`	`@"S:name[,name2]…"[,g]`

Scratches one or more files, also with the wildcards `*` and `?`. The message does not appear by itself; `@` afterwards shows how many files were scratched, e.g. `01, FILES SCRATCHED,02,00` for two files.

Example	Effect
`@S:TEST`	scratch the file TEST
`@S:*`	scratch all files on the disk
`@S:T*`	all files beginning with T
`@S:?T`	all files with two characters and T in second place
`@S:BITS,BYTES,WORDS`	scratch the three files

4.22 Locking and unlocking files

Direct mode	Program mode
`@L:name`	`@"L:name"[,g]`

Toggles the write protection of a file: a locked file cannot be scratched by accident and shows a `<` after the type in the directory (e.g. `PRG<`). Another `@L` removes the lock. Does not work with 1581 or MSD drives without JiffyDOS.

4.23 Output to the printer

Direct mode	Program mode
`@P[,g][,sa]`	`@P[,g][,sa]`

Redirects screen output (LIST, directory, `@D`, `@T`) to the printer; a second `@P` switches back. The default is device 4 with secondary address 0 (upper case/graphics); secondary address 7 prints upper and lower case.

Example	Effect
`@P`	output to printer 4, secondary address 0 – or back to the screen
`@P,5`	output to printer 5
`@P,5,7`	output to printer 5, upper/lower case

4.24 Printing the screen

`<CTRL>+<P>` prints the current text screen in the matching character set. The printer must be device 4.

4.25 Turning function keys off and on

Direct mode	Program mode
`@F`	`@F`

Turns off the JiffyDOS function keys (in 128 mode the C128 assignment then applies). Back on with `SYS 58551` (JiffyDOS/64: commands and function keys) or `SYS 65137` (JiffyDOS/128).

4.26 Turning JiffyDOS commands off and on

Direct mode	Program mode
`@Q`	`@Q`

Turns off all JiffyDOS commands and function keys and restores the original BASIC vectors. Fast transfer to the drive stays active.

Back on with `SYS 58451` (JiffyDOS/64, commands only, without function keys) or `SYS 65137` (JiffyDOS/128). This is also needed when the JiffyDOS kernal was switched in with the computer on, or when a program has changed the BASIC vectors.

5 DOS Wedge enhancements

5.1 Direct and program mode

In program mode, command and file name usually have to be in quotes: `@S:NAME` only works directly, in a program it is `@"S:NAME"`.

Program line	Effect
`10 @"$:*"`	Show the directory
`20 @"S0:NAME"`	Scratch a file
`30 @#9`	Default device 9
`40 @"D:NAME"`	List a BASIC program from disk
`50 @"I:"`	Initialise the default drive

5.2 Command chaining

In program mode JiffyDOS commands can be combined with BASIC and other JiffyDOS commands on one line. In direct mode this works to a limited extent if the program form is used and no BASIC command is in between.

Program line	Effect
`10 @#9:@"S:TEST":@#8`	change to device 9, scratch TEST, back to device 8
`20 @"$:*":PRINT:@`	directory, empty line, error channel

5.3 String variables

Unlike the Commodore wedge, commands may contain string variables. This makes short interactive utilities possible without hard-coded file names:

```
10 REM SCRATCH A FILE
20 @"$"
30 INPUT"FILENAME";N$
40 @"I:"
50 @"S:"+N$
60 END
```

5.4 Another device for one command

A device at the end of a command applies to that command only; the default device stays. For this the command must be in quotes and the number must follow a comma. This works in direct and program mode, also with a variable:

Command	Effect
`@"S0:NAME",9`	Scratch a file on device 9
`@"$:*",9`	Directory of device 9
`@"",10`	Error channel of device 10

Command	Effect
`10 A=9:@"$",A`	Directory of device 9 via a variable

6 Command summary

6.1 DOS Wedge 5.1 commands

Command	Effect
`@`	Read the error channel
`@C:neu=alt`	Duplicate a file on the same disk
`@I`	Initialise the drive
`@N:diskname,id`	Format a disk
`@N:diskname`	Short format
`@Q`	Turn off JiffyDOS commands
`@R:neu=alt`	Rename a file
`@S:name[,name2]…`	Scratch file(s)
`@UJ`	Reset the drive
`@V`	Validate a disk
`@$`	Show the directory
`@#g`	Set the default device
`/name`	Load a BASIC program
`↑name`	Load and run a BASIC program
`%name`	Load a machine-language program
`←name`	Save a BASIC program

6.2 Additional JiffyDOS commands

Command	Effect
`@B`	Turn off 1541 head knocking
`@D:name`	List a BASIC program from disk
`@F`	Turn off function keys
`@Gwert`	Set the sector interleave
`@L:name`	Lock/unlock a file
`@O`	Un-NEW
`@P`	Printer output on/off
`@T:name`	List a text file from disk
`@Xg`	Destination drive for the copier
`*"name"typ`	Copy a file
`£name`	Load and run a machine-language program
`'name`	Verify a file
`@/0:name`, `@/`	Change into a 1581 partition, back to the root directory
`<CTRL>+<A>`	Mark/unmark all files for copying

Command	Effect
<CTRL>+<D>	Change the default drive
<CTRL>+<P>	Print the screen
<CTRL>+<S>, <CTRL>+<W>	Pause listing, step line by line
<CTRL>+<W>	In the loaded directory: mark/unmark a file for copying
<SHIFT>+<RUN/STOP>	Load and run the first program on the disk
SYS 58451	Turn JiffyDOS commands back on (64 mode)
SYS 58551	Turn commands and function keys back on (64 mode)
SYS 65137	Turn commands and function keys back on (128 mode)

7 Using the DolphinDos26 kernal

7.1 The 1541 Switchless Multi-Speeder V3

The **1541/1571 Switchless 8x Multi-Floppy-Speeder V3** is an expansion that is fitted into the CPU socket of the 1541 or of the 1571 in the C128DCR. It replaces the DOS ROMs with a 512 KB EPROM holding **eight DOS banks** that are changed by DOS command without a switch – on the 1541 among them JiffyDOS in bank 4 and DolphinDos26 in bank 8. It also adds **512 KB of RAM** in the drive and a 6821 PIA as an 8-bit parallel port to the C64 (DolphinDOS cable on the user port).

DolphinDos26 is the drive firmware for this hardware, a further development of DolphinDOS 3.0. It transfers data in parallel instead of serially and uses the large memory, e.g. to read whole disks into the drive RAM and write them back to any number of disks, or as a RAM disk.

7.2 What the DolphinDos26 kernal does for the computer

The C64 kernal **DolphinDos26** (`C64DolphinDos26.bin`, 8 KB; for the C64 II with its combined BASIC/kernal chip `C64IIDolphinDos26.bin`, 16 KB) replaces the computer's kernal. It combines three transfer methods and picks the fastest one the addressed device supports on every access:

1. **DolphinDOS parallel** over the parallel cable when the drive speaks DolphinDos26 or DolphinDOS (shown as `LOADING DD:`),
2. **JiffyDOS** when the device reports itself as a JiffyDOS device – e.g. a 1541 with JiffyDOS (bank 4 of the Multi-Speeder) or an SD2IEC (shown as `LOADING JF:`),
3. **serial** like the original kernal for all other devices, such as printers or drives without a fast loader.

So **one single kernal** handles every device on the bus; no kernal switch is needed for the transfer. On top come the DolphinDOS commands with `@`, the X commands for DolphinDOS drives and function keys for navigating on the SD2IEC.

	JiffyDOS kernal 6.01	DolphinDos26 kernal
Fast transfer	JiffyDOS over the serial bus	DolphinDOS parallel, JiffyDOS or serial – automatically per device
Extra cable	none	parallel cable on the user port for DolphinDOS drives; none for JiffyDOS devices
Commands	JiffyDOS commands of this manual, file copier	DolphinDOS commands (@, X commands), SD2IEC function keys
Licence	commercial, from RETRO Innovations	free for private use; contains JiffyDOS transfer routines (see Copyright)

A measurement with 16,000 bytes shows how much the parallel transfer gains:

Operation	Standard 1541	JiffyDOS 1541	DolphinDos26 parallel
LOAD	41.2 s	6.7 s	2.1 s
SAVE	48.4 s	19.1 s	4.6 s

7.3 Advantages on the Commodore 64 Ultimate

The **Commodore 64 Ultimate (C64U)** emulates its 1541 including a freely selectable drive ROM and a parallel cable on the drive's VIA1. For this there is the drive firmware **DolphinDos26UL** (`C1541_DOLPHINDOS26UL.bin`). Together with the DolphinDos26 kernal this gives these advantages:

- **No soldering, no EPROM:** kernal and drive ROM are selected in the C64U menu and can be reset at any time.
- **Parallel instead of serial:** the emulated drive loads and saves over the parallel cable – in the measurement above about three times as fast loading and four times as fast saving as with JiffyDOS.
- **One kernal for everything:** a connected SD2IEC or a real JiffyDOS 1541 is addressed via JiffyDOS with the same kernal, printers and other devices serially – without changing the kernal.
- **DolphinDOS convenience:** directory without losing memory, disk commands with `@`, track cache in the drive and the MS-Tools (with the C64U's REU).

> **Note:** The DolphinDos26 kernal contains only the JiffyDOS **transfer**. The commands, function keys and file copier of this manual are available only with the original JiffyDOS kernal.

7.4 More information

ROM files, the manuals for DolphinDos26 (Multi-Speeder) and DolphinDos26UL (C64U and DolphinDOS 2 hardware), the MS-Tools and the technical notes are on GitHub: github.com/FraEgg/DolphinDOS26

8 Using SD2IEC

8.1 What is SD2IEC?

SD2IEC is firmware by Ingo Korb for small devices that make a memory card appear as a disk drive on the serial bus. Such SD card readers are sold under various names or can be built yourself, e.g. uIEC. The files are on a FAT-formatted card, individually or in disk images (`.D64`, `.D41`, `.D71`, `.D81`, `.DNP`), and can be organised in directories.

JiffyDOS support is already built natively into all sd2iec-based drives. The firmware detects the protocol and delivers the files accordingly. So neither a JiffyDOS ROM nor a setting is needed on the device. On the computer the JiffyDOS kernal is all you need.

> **Commodore 64 Ultimate:** The C64U also offers SD2IEC, with no additional device; see the section “On the Commodore 64 Ultimate”.

8.2 Sending commands correctly

JiffyDOS sends everything after `@` to the default device – with one exception: if one of the letters D, L, T, B, F, O, P, Q, X, G or the character # directly follows the `@`, JiffyDOS runs a command of its own. SD2IEC commands beginning with one of these letters must therefore be in **quotes**. Commands with C, M, R, S, N, U and V work directly.

Input	Result
`@CD:GAMES`	correct: goes to the device
`@"X?"`, `@"T-RA"`, `@"G-P"`	correct: goes to the device
`@XW`	wrong: JiffyDOS treats it as its own command, the device receives nothing
`@T-RA`	wrong: JiffyDOS looks for a text file named -RA
`@G-P`	wrong: `?SYNTAX ERROR`

Usually the SD2IEC is not the only drive. `@#9` or `<CTRL>+<D>` makes it the default device; for a single command the device number at the end is enough, e.g. `@"CD:GAMES",9`.

8.3 Changing directories

Command	Effect
`@CD:GAMES`	change into the subdirectory GAMES
`@CD/GAMES`	the same
`@CD←`	one directory back
`@CD:←`	the same
`@CD//`	change to the root directory of the card
`@CD//GAMES`	change to GAMES in the root directory, wherever you are
`@CD/GAMES/:ARCADE`	change into the directory ARCADE inside GAMES

Command	Effect
`@CD/GAMES/ARCADE`	the same
`@CD:GAM*`	change into the first matching directory; wildcards are allowed anywhere in the path
`@MD:NEW`	create the directory NEW
`@MD/GAMES/:NEW`	create the directory NEW inside GAMES
`@RD:NEW`	delete the directory NEW (only subdirectories of the current directory)

The character `←` is the arrow key at the top left of the C64 keyboard. Drive and partition numbers are ignored by these commands.

8.4 Using disk images

An image is entered like a directory. The SD2IEC then behaves as if this disk were in the drive; `@$`, `/`, `%` and the other commands work with the image. The image must be the last part of the path and follow the colon. It is recognised by file extension and size.

Command	Effect
`@CD:GAME.D64`	insert an image from the current directory
`@CD/GAMES/:GAME.D64`	insert an image from a subdirectory
`@CD←`	leave the image, back to the directory
`@"XS:LIST.LST"`	activate a swap list for programs with several disks; the first image is inserted
`@"XS"`	turn the swap list off

A swap list is a text file with one image name per line. The disk change button on the device moves to the next image. If no list is set, the device uses the file `AUTOSWAP.LST` in the current directory.

8.5 Directory and partitions

`@$` and F1 work as usual. Directories appear with the type `DIR`. Long file names are supported; if a name is longer than 16 characters, the device shows the short 8.3 name.

Command	Effect
`@$:*=D`	show directories only
`@$:*=H`	also show hidden files
`@$=T`	directory with date and time
`@$=P`	show partitions
`@CP2`	select partition 2 as the current partition

8.6 Further commands

Command	Effect
`@S:NAME`	scratch a file; wildcards and comma-separated lists are allowed, directories are skipped
`@R:NEW=OLD`	rename a file

Command	Effect
`@C:COPY=ORIGINAL`	duplicate a file or append files
`@C/BACKUP/:COPY=ORIGINAL`	put the copy in another directory
`@N:NAME,ID`	format; only with a D64 image inserted
`@UJ`	close channels; directory, image and swap list are kept
`@"X"`	show the device settings
`@"XW"`	store the settings and device number permanently
`@"T-RA"`	read the time (only with a built-in real-time clock)

The device number is changed with a short command and then stored permanently, here from 8 to 9. If the device has switches or solder jumpers for the number, they should match.

```
OPEN 15,8,15,"U0>"+CHR$(9):CLOSE 15
@#9
@"XW"
```

8.7 Notes and limits

- An SD2IEC does **not run 1541 code**. Programs with their own fast loader or copy protection in the drive therefore do not work; only some well-known fast loaders are reproduced in the firmware, e.g. Turbodisk, Final Cartridge III, EXOS, Epyx FastLoad, Dreamload and GEOS.
- `@G` and `@B` have no function, because an SD2IEC has neither a sector interleave nor a read/write head. `@L` is not among the documented SD2IEC commands.
- The file copier also works between an SD2IEC and another drive, because it uses normal file access. It copies individual files, not directories.

8.8 With the DolphinDos26 kernal

With the DolphinDos26 C64 kernal (`C64DolphinDos26.bin`), an SD2IEC is also addressed via JiffyDOS, shown by `LOADING JF:`. The JiffyDOS commands of this manual are not available there, but the drive commands are. You set the default device with `@9`; it applies until the next reset. Everything after the `@` goes to the device unchanged, so quotes as with JiffyDOS are not needed.



Figure 4: Directory of an SD2IEC: directory, disk image and files



Figure 5: Changing directory and inserting a disk image with the DolphinDos26 kernal

Three function keys make the work easier. For C=+F5 and C=+F7, first show the directory with `@$` and move the cursor to the desired line:

Key	Writes	Effect
C=+F3	`@CD←`	one directory or image back
C=+F5	`"CD:` before the name	change into the directory or image in the cursor line
C=+F7	`LOAD"name":`	prepare the file in the cursor line for loading; RETURN loads it to the address in the file, from the device set with `@9`, otherwise device 8



Figure 6: C=+F7 on a directory line, then RETURN: loading via JiffyDOS

8.9 On the Commodore 64 Ultimate

On the C64U, in the menu under `SoftIEC Drive Settings`, the item `IEC Drive` turns the drive on, `Soft Drive Bus ID` sets the device number and `Default Path` the start directory, e.g. `/USB1/SD2IEC/`. The drive reports itself as `73,U64IEC ULTIMATE DOS V1.1`. It is a separate reproduction of SD2IEC and does not know all commands. With C64U firmware 1.1.0 the following applies:

Command	Effect on the C64U
`@CD:GAMES`	change into the subdirectory GAMES
`@CD:GAMES/ARCADE`	several levels at once
`@CD:..`	one level back
`@CD←`	back to the start directory (`Default Path`), not just one level
`@CD//`	to the top level of the C64U
`@CD:/USB1/SD2IEC`	full path
`@CD:GAME.D64`	insert a disk image; leave with `@CD←`
`@MD:NEW`, `@RD:NEW`	create and delete a directory, only in the current directory
`@S:NAME`, `@R:NEW=OLD`, `@C:COPY=ORIGINAL`, `@I`	as with the SD2IEC

Not available are the forms `CD/name` and `MD/…`, the X commands, and `T-R`, `G-P`, `CP`, `UJ`, `V`, `N` and `L`. The drive remembers its directory across a reset of the C64. The files are on the USB stick or in the configured directory. Measured with 16 KB: LOAD 1.8 s, SAVE 5.5 s.

More information: www.sd2iec.de

9 Using Pi1541 and Pi1541-II

9.1 What is Pi1541?

Pi1541 by Stephen White turns a Raspberry Pi into a cycle-exact emulated 1541. Unlike an SD2IEC, it reproduces the drive's 6502 processor and both 6522s and runs the real DOS ROM and any drive code. That is why fast loaders and many copy-protected originals work on it. The disks are images on an SD card, in the folder `1541`.

For JiffyDOS this means: the Pi1541 has **no built-in JiffyDOS**. It becomes as fast as a JiffyDOS 1541 once it is given the JiffyDOS 1541 ROM as its drive ROM.

The **Pi1541-II** is separate hardware for the same software: a module for the Raspberry Pi 3B with a 7406 driver (circuit variant “Option B”), OLED display, five buttons, buzzer and two serial sockets. Everything in this chapter applies to it for JiffyDOS; in addition there are the settings in the section “Settings for the Pi1541-II”.

9.2 Setting up the JiffyDOS ROM

1. Copy the JiffyDOS 1541 ROM (16 KB, licence required) to the **root directory** of the SD card, not to the folder `1541`.
2. Enter the ROMs in the file `options.txt`; up to seven are possible (see below).
3. At start-up `ROM1` is active. You choose other ROMs in browser mode with F1 to F7 on a USB keyboard or on the device: hold button 1 and press one of the buttons 2 to 5.
4. Activate the JiffyDOS kernal on the C64, insert an image and check with `@`: the start-up message is `73,JIFFYDOS 5.0 1541,00,00`.

```
ROM1 = JiffyDOS_C1541.bin
ROM2 = dos1541
deviceID = 8
```

Without an entry the Pi1541 looks in the root directory for a file named `d1541.rom`, `dos1541`, `d1541II` or `Jiffy.bin` and uses the first one as `ROM1`. The original ROM as `ROM2` is worthwhile for programs that do not run with JiffyDOS.

9.3 Settings for the Pi1541-II

The Pi1541-II also needs these entries in `options.txt`. The line `LCDName` must match the fitted display; some modules need `sh1106_128x64`.

```
splitIECLines = 1
LCDName = ssd1306_128x64
i2cBusMaster = 1
i2cLcdAddress = 60
SoundOnGPIO = 1
LcdLogoName = 1541ii
```

9.4 The two operating modes

Mode	What happens	Meaning for JiffyDOS
Browser mode (also SD2IEC mode)	after start-up; the Pi1541 answers simple commands itself: changing directories, choosing images, loading files	the commands in the table below; the JiffyDOS ROM is not yet in operation here
Emulation	after inserting an image; a complete 1541 runs with the chosen ROM	full JiffyDOS speed and all commands of this manual, including `@G`, `@B` and `@L`

9.5 Directories and images from the C64

In browser mode the same forms apply as with the SD2IEC:

Command	Effect
`@CD:GAMES`	change into the subdirectory GAMES
`@CD←`	one directory back
`@CD//GAMES`	change to GAMES in the top directory
`@CD/GAMES/:ARCADE`	change into the directory ARCADE inside GAMES
`@CD:GAME.D64`	insert an image and start the emulation
`@MD:NEW`	create a directory
`@RD:NEW`	delete a directory
`@N:NAME`	create a new D64 image
`@S:NAME`	scratch a file
`@R:NEW=OLD`	rename a file

Partitions, sector commands such as `U1` and the SD2IEC X commands are not available in browser mode. The CBM FileBrowser that comes with the Pi1541 is often more convenient: if it is the first file in the folder `1541`, `LOAD"*",8` starts it.

During emulation the real 1541 ROM answers, and it does not know `@CD`. So you leave an inserted image on the Pi1541 itself: with the “Exit Folder” button or with ESC on the USB keyboard.

9.6 Operating the device

Button	Effect
Select/Reset	select in the browser, insert an image
Previous/Up, Next/Down	move in the browser; during emulation previous or next disk of a selection
Exit Folder	leave a folder or image
Insert	add an image to the selection for several disks
Hold Insert and button 1 to 4	set device number 8 to 11

USB keyboard	Effect
ENTER	insert an image and start the emulation

USB keyboard	Effect
INSERT	add an image to the selection; during emulation the number keys change the disk
ESC	leave the image and emulation
BACKSPACE	leave a folder
ALT+N	create a new D64 image
ALT+W	toggle the write protection of the image
F1 to F7	choose the drive ROM
F8 to F11	set device number 8 to 11

9.7 Notes and limits

- Writing is only possible to D64 and G64 images. Changed images are written back when you leave the emulation; do not switch off the Pi1541 until it shows that it is safe.
- The simple circuit “Option A” is only suitable if no other device is on the serial bus. The Pi1541-II is “Option B” and can be used together with other devices.
- The emulated 1541 is not always reset when the C64 is reset. Remedy: leave the image and insert it again.
- Programs that do not run with JiffyDOS are started with the original ROM (e.g. `ROM2`) and, if necessary, the original C64 kernal.
- The Pi1541 is connected only to the serial bus. With the DolphinDos26 C64 kernal the transfer therefore runs via JiffyDOS (with the JiffyDOS ROM) or serially, not in parallel.

More information: cbm-pi1541.firebaseapp.com and for the Pi1541-II github.com/Tebl/C64-Pi1541-II

10 Appendix

10.1 1581 partitions

With the wedge you change into a partition (subdirectory) on the 1581 that is visible in the current directory, and back to the root directory. Before copying from or into a partition, change into it first.

Effect	Direct mode	Program mode
change into partition name	`@/0:name`	`@"/0:name"[,g]`
back to the root directory	`@/`	`@"/"[,g]`

10.2 Super Graphics Gold printer interface

The Xetec Super Graphics Gold interface answers burst commands on device 14; `<CTRL>+<D>` therefore also shows 14. Do not send JiffyDOS commands to device 14, or the bus hangs. With a JiffyDOS 1571, hangs occur more often; the remedy after switching on (and after pressing RESET and CLEAR on the interface together) is:

```
OPEN 15,4,15:PRINT#15,"SL":CLOSE 15
```

10.3 Memory locations of the JiffyDOS kernal (C64)

Addresses of the JiffyDOS 6.01 kernal that are useful for your own programs:

Address	Meaning
155 ($9B)	Function keys: 0 = JiffyDOS assignment active
176/177 ($B0/$B1)	Pointer to the function key texts
190 ($BE)	Default device of the JiffyDOS commands
58451 ($E453)	Turn on JiffyDOS commands (BASIC vectors)
58551 ($E4B7)	Turn on commands and function keys

10.4 Origin and acknowledgements

Project	Developers	Contribution
JiffyDOS	Mark Fellows; Creative Micro Designs (CMD), today Jim Brain, RETRO Innovations	Operating system, original manual V6.01
Commented ROM listing	Magnus, revision B, 1996	Memory locations and command table of the JiffyDOS 6.01 kernal
DolphinDOS 3.0	Günther Jilg and Jan Bubela, Dolphin Software	Basis of DolphinDos26
VICE	the VICE team	Emulator for tests and figures

Special thanks go to Jim Brain of RETRO Innovations for his support with the JiffyDOS implementation of the C64/C128 kernal DolphinDos26.

New edition of this manual: **Frank Eggen**.

Topic	Address
Contact	frankeggen@emden.net
Projects on GitHub	https://github.com/fraegg
DolphinDOS26: firmware, kernals, MS-Tools, manuals	github.com/FraEgg/DolphinDOS26
Switchless Multi-Speeder for the 1541 (hardware)	github.com/FraEgg/commodore-1541-switchless-floppydrive-8x-multi-floppy-speeder
JiffyDOS ROMs and licences	www.jiffydos.com

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CBM DOS 1541, C64 kernal and BASIC	© Commodore

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